

Lesson : Flexbox in CSS

Module 9 : Notes

Introduction to CSS Flexbox

- Flexbox (Flexible Box):
 - Designed for UIs, aligns and distributes space among items.
 - Ensures consistent layout across screen sizes.
- Axes Understanding:
 - Main axis: Primary layout direction (horizontal or vertical).
 - Cross axis: Perpendicular to main axis, determines alignment.
- Usage:
 - Apply display: flex; to enable Flexbox.
 - Default: Horizontal main axis, vertical cross axis.
- Direction:
 - flex-direction: row;; Horizontal main axis.
 - flex-direction: column;; Vertical main axis.
- Cross Axis:
 - Determined by perpendicular axis to flex-direction.
 - Horizontal cross axis for flex-direction: row;;, vertical for flex-direction: column;.

Flex container properties

- **Align Items**
 - Aligns flex items along the cross-axis.
 - Possible values: stretch, center, flex-start, flex-end.
 - The default value is stretch.
- **Align Content**
 - Aligns the flex lines within a flex container when there is extra space on the cross-axis.
 - Possible values: **stretch, center, flex-start, flex-end, space-between, space-around**.
 - Used when there are multiple lines of flex items.
- **Justify Items**
 - Aligns individual items along the main axis within their alignment container.
 - Possible values: **auto, normal, start, end, center, stretch**.
 - Influences the distribution of space around content items.
- **Justify Content**
 - Aligns the entire flex container's items along the main axis.
 - Possible values: **flex-start, flex-end, center, space-between, space-around, space-evenly**.
 - Determines how space is distributed around items.
- **Gap**
 - Sets the spacing between items in a flex container.
 - Can be defined for rows (row-gap) and columns (column-gap), or both (gap).
 - Useful for creating consistent spacing without margins.
- **Flex Wrap**

- Controls whether flex items are forced into a single line or can wrap onto multiple lines.
- Possible values: **nowrap**, **wrap**, **wrap-reverse**.
- The default value is **nowrap**.
- **Flex Flow**
 - Shorthand for setting both flex-direction and flex-wrap.
 - Values are space-separated: flex-direction flex-wrap.
 - Defines both the direction and wrapping behavior of flex items.
- **Flex Direction**
 - Specifies the direction of the flex items in the flex container.
 - Possible values: row, row-reverse, column, column-reverse.
 - Determines the main axis direction (horizontal or vertical).

Flex item :

- A child element of a flex container.
- Inherits all the properties and behaviors of flexbox layout.
- Can be manipulated individually using flex properties (order, flex-grow, flex-shrink, flex-basis.).
- **order**
 - Defines the item's order within the container.
 - Default: 0.
- **flex-grow**
 - Determines how much the item can grow.
 - Default: 0 (no growth).
- **flex-shrink**
 - Determines how much the item can shrink.
 - Default: 1.
- **flex-basis**
 - Sets the initial size of the item.
 - Default: auto.
- **flex**
 - Shorthand for **flex-grow**, **flex-shrink**, **flex-basis**.
 - Default: 0 1 auto.
- **align-self**
 - Overrides **align-items** for an individual item.
 - Possible values: **auto**, **flex-start**, **flex-end**, **center**, **baseline**, **stretch**.